# Sidharth Singh

Backend engineer building high-throughput distributed systems and AI-powered platforms

sidharthsingh7111@gmail.com Delhi, India GitHub: sid563 LinkedIn: sidharthsingh-ss563

## SUMMARY

Backend-leaning engineer with 2+ years of production experience building high-throughput microservices, event-driven pipelines and AI-assisted workflows for procurement, logistics and supply-chain platforms. Rust (Actix Web, Tokio) and TypeScript (NestJS) for services, Python (FastAPI) for data-heavy workflows, Kafka for everything asynchronous, PostgreSQL and MongoDB at 100M+ document scale. Owns a feature end to end, from marketplace or carrier integration through to production, and treats AI coding tools as part of design, implementation and review.

## EXPERIENCE

Mar 2026 – Present
Okhla, Delhi

### Software Engineer

1buy.ai · Sharpbuy Global Solutions Pvt. Ltd.

AI-powered B2B electronic component intelligence, sourcing and procurement platform.

- Building the order-fulfilment backend in Python (FastAPI), PostgreSQL and Kafka: purchase order to shipment to customs workflows with DHL carrier API integration, AI parsing of trade documents (packing lists, Bills of Entry) via Claude, and event-driven cross-department messaging.
- Designed the inbound-mail pipeline: cheapest-first anchoring of every email to its order or shipment, a two-stage classify-then-act flow over a plug-in registry of 16 LLM handlers, per-step model routing, and dispatch through a transactional outbox, with deterministic no-network tests.
- Design and build end-to-end backend services on a TypeScript, NestJS and MongoDB stack with hexagonal architecture, on an ultra-small engineering team.
- Engineered MongoDB aggregation pipelines and compound-indexing strategies over 100M+ pricing documents to power real-time component discovery.
- Rewrote the background job scheduler in Rust on Tokio: Arc<Semaphore>-bounded worker pools, atomic job claims and cancellation-safe tasks for concurrent, multi-instance processing without thread contention.
- Wrote shared pricing and formula logic once in Rust and compiled it to WebAssembly with wasm-bindgen, passing batches across a columnar Float64Array boundary, so the TypeScript and Rust codebases share one source of truth with parity tests on both sides.
- Developed automated RFQ ingestion pipelines and multi-currency calculation engines with Decimal128 precision, integrating LLM APIs into sourcing workflows.

`Python · FastAPI · PostgreSQL · Kafka · TypeScript · NestJS · MongoDB · Rust · Tokio · WebAssembly · Claude API`

Apr 2025 – Feb 2026
Gurugram

### Software Engineer

RapidShyp · Om Logistics Limited

B2B logistics and shipping platform.

- Engineered core backend microservices in Rust (Actix Web on the Tokio runtime) and Java: async handlers, serde-typed domain models and backpressure-aware worker pools, improving API concurrency and throughput while reducing latency under peak shipping loads.
- Built event-driven pipelines on Apache Kafka for real-time order processing, shipment status synchronisation and multi-seller fulfilment across marketplace and storefront sales channels.
- Integrated carrier and logistics partner APIs for shipment creation, tracking and status updates, so sellers on multiple sales channels ship through one platform.
- Integrated Redis for low-latency caching alongside TiKV and etcd for distributed key-value storage, service coordination and configuration management.
- Optimised schema design and query execution across MongoDB and MySQL for high-volume logistics read and write paths.
- Won first place across all engineering teams in the RapidShyp internal hackathon (Mar 2025) and the Customer Obsession Award (May 2025) for feature ownership, rapid execution and reliable production delivery.

`Rust · Actix Web · Java · Kafka · MongoDB · MySQL · Redis · TiKV · etcd`

Jul 2024 – Mar 2025
Gurugram

### Associate Software Engineer

RapidShyp · Om Logistics Limited

- Designed and owned the B2B Bulk Orders and Purchase Requisition flows in the Order Management System: bulk upload and validation, multi-seller shipment creation, approval workflows and order orchestration at scale.

- Brought marketplace and multi-channel seller orders into one fulfilment pipeline, so every sales channel flows through the same order lifecycle and logistics integrations.
- Developed order-lifecycle dashboards and the internal Warehouse Management System frontend in React and Next.js, used daily by operations teams.
- Contributed to RESTful microservices in Rust (Actix Web) and Java, integrating third-party logistics carrier APIs.
- Containerised local development environments with Docker and optimised query patterns across MongoDB and MySQL.

Rust · Java · React · Next.js · MongoDB · MySQL · Docker

## EDUCATION

Sept 2020 – Jun 2024

**B.Tech, Computer Science & Engineering (Bioinformatics)**
Vellore Institute of Technology (VIT), Vellore
CGPA 8.22 / 10

## SKILLS

| | |
|---|---|
| **Languages** | Rust · TypeScript · JavaScript · Python · Java · SQL · Go |
| **Backend & APIs** | Actix Web · Tokio · NestJS · FastAPI · Spring Boot · REST APIs · gRPC · WebSockets · Hexagonal architecture · Microservices |
| **Data & messaging** | PostgreSQL · MongoDB · MySQL · Redis · Apache Kafka · etcd · TiKV |
| **Infra & cloud** | Docker · Kubernetes · AWS (EC2, S3, Lambda) · CI/CD · Linux |
| **AI & search** | LLM integration (Claude) · RAG · Atlas Vector Search · pgvector · AI document parsing |
| **Frontend** | React · Next.js |

## AWARDS & CERTIFICATIONS

### Awards

2025 — **First place, RapidShyp internal hackathon 2025**
Across all engineering teams, for an innovative prototype built under tight time constraints (Mar 2025)

2025 — **Customer Obsession Award, RapidShyp**
For exceptional feature ownership, rapid execution, going the extra mile and reliable production delivery (May 2025)

### Certifications

2023 — **Google Cloud Computing Foundations**
Google & VIT

2023 — **Modern Application Development with Java Spring Boot**
VMware IT Academy

2023 — **Core Technical Skills (Docker, Kubernetes, MongoDB)**
VMware IT Academy

## SELECTED PROJECTS

**D-MPNN drug property predictor**

A directed message-passing neural network with ensemble learning for ADME property prediction, trained on 12,859 molecular SMILES entries with substructure features.

Python · PyTorch · Cheminformatics · Ensemble learning

**Personal RAG assistant on AWS**

A retrieval-augmented chat service built to learn the full build-and-deploy loop: NestJS, MongoDB Atlas vector search, Hugging Face embeddings, Groq-hosted Llama, Docker, and a hand-provisioned EC2 box.

NestJS · TypeScript · MongoDB Atlas · Hugging Face · Groq · Docker